

Robust Nonlinear Control for Boost Converter

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Abstract—The standard current loop is modified for a boost converter to eliminate sensitivity of the control-to-output transfer function to the nature and magnitude of resistive load. An additional term directly proportional to the load current and output voltage, and inversely proportional to the input voltage is added to the current loop. This results in practically invariant loop gains for different resistive loads, including constant power load.

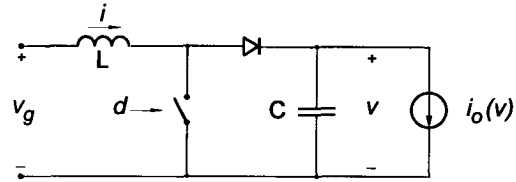


Fig. 1. Idealized dc-dc boost converter.

I. INTRODUCTION

IN many applications switching power converters feed time-varying nonlinear loads or other converters. In those cases, performance of the converters with conventionally designed control loops may not be satisfactory.

This paper concentrates on control of a boost dc-dc converter loaded by unknown (linear or nonlinear) resistive load. Besides the linear resistor, a resistive load of particular interest is the constant power load (CPL), which is often used as a simplified model for the input impedance of a converter with output voltage regulation. It was shown [1] that a boost converter loaded by a linear resistor is open-loop stable, and one loaded by CPL is open-loop unstable. The stability can be achieved in both cases by closing the current loop [1]. However, the control-to-output transfer functions differ significantly when the current loop is closed. For that reason, a voltage compensator that would provide optimum performance for both linear resistive load and CPL cannot be designed. In the nonlinear control algorithm proposed in this paper, the current loop is modified so that its sensitivity to the nature and magnitude of the resistive load is reduced.

Section II presents derivation of the modified current-mode control for a boost converter. Small-signal characteristics of the control are analyzed in Section III, and compared to the characteristics of conventional current-mode control. Complete small-signal model for the modified algorithm is given in Section IV. Invariant properties of the modified current-mode control are confirmed experimentally, and the results are shown in Section V. Output voltage responses to large step change of the load current are shown for conventional and modified current-mode control laws. Application of the modified current-mode control to the buck and the flyback converter is discussed in Section VI.

II. DERIVATION OF MODIFIED CURRENT LOOP

The average model of an idealized boost dc-dc converter (Fig. 1) in the continuous conduction mode is given by [2]:

$$\begin{aligned} \frac{di}{dt} &= \frac{v_g}{L} - \frac{d'}{L}v, \\ \frac{dv}{dt} &= \frac{1}{C}d'i - \frac{1}{C}i_o(v) \end{aligned} \quad (1)$$

where $d' = 1 - d$, and $i_o = i_o(v)$ is the load resistor characteristic. A linear resistor is given by $i_o = v/R$, while CPL is described by $i_o = P/v$, where P is the output power. For the conventional current mode control [3] (current injection control (CIC) or standard control module (SCM)), which will be abbreviated in the text as CCMC, the average duty ratio is

$$d = -FM(R_i i - v_c) \quad (2)$$

where FM is the modulator gain, R_i is the current sense resistor, and v_c is the output of the voltage compensator, i.e. the modulator control input. Expression (2) is valid at frequencies significantly lower than the switching frequency. The small-signal model for $v_g = V_g = \text{const.}$ can be obtained from (1) and (2) by linearization about the operating point $V = V_g/D'$, and $I = I_o/D'$, and is given by:

$$\begin{aligned} \frac{d\hat{i}}{dt} &= -\frac{D'}{L}\hat{v} + \frac{V}{L}\hat{d}, \\ \frac{d\hat{v}}{dt} &= \frac{D'}{C}\hat{i} - \frac{1}{C}\frac{di_o}{dv}\hat{v} - \frac{I_o}{CD'}\hat{d} \end{aligned} \quad (3)$$

where the $(\hat{\cdot})$ denotes the perturbed variable and uppercase letters denote operating point values. The perturbed duty ratio for CCMC from (2) is

$$\hat{d} = \frac{\partial d}{\partial i}\hat{i} + \frac{\partial d}{\partial v_c}\hat{v}_c = -FM R_i \hat{i} + FM \hat{v}_c. \quad (4)$$

Control-to-output transfer function, $G_c = \hat{v}/\hat{v}_c$, can be readily derived from (3) and (4). The transfer functions shown in Fig. 2 were measured on the experimental boost converter with parameter values given in Appendix A, for the different values of the linear resistive load and CPL. The measurements illustrate theoretically expected significant dependence of the

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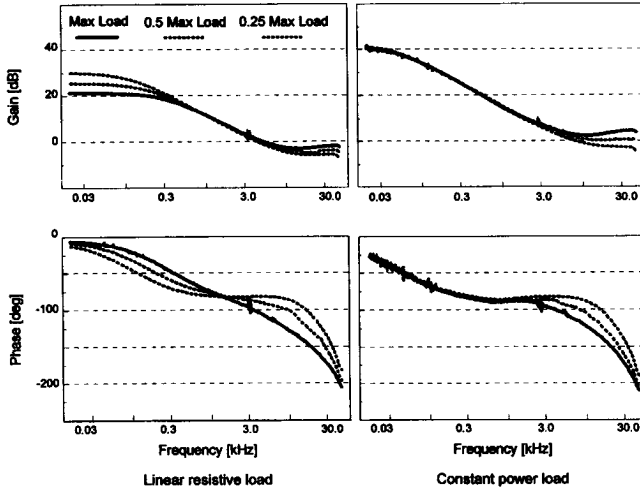


Fig. 2. CCMC control-to-output transfer functions.

transfer function on the load variation. In particular, the transfer function dc gain changes noticeably with the load type, and for linear resistive load it depends on the load current. Although a voltage controller can be designed to assure small-signal stability in all cases, the converter performance cannot be optimal for both load types and any load current.

A possible solution is to modify the current loop to achieve constant dc gain of the transfer function G_c in all cases. The simplest modification is to add a term which carries information about the load current in the current loop. Then, the average duty ratio is given by

$$d = -FM(R_i i - v_c - f(i_o(v), v, v_g)) \quad (5)$$

where $f(i_o, v, v_g)$ is a function which should be determined so that the dc gain becomes independent of load.

From (5), if the input voltage v_g is constant, the perturbed duty ratio is

$$\hat{d} = \frac{\partial d}{\partial i} \hat{i} + \frac{\partial d}{\partial v} \hat{v} + \frac{\partial d}{\partial v_c} \hat{v}_c \quad (6)$$

where now

$$\begin{aligned} \frac{\partial d}{\partial i} &= -FM R_i, \\ \frac{\partial d}{\partial v} &= FM \frac{df}{dv} = FM \left(\frac{\partial f}{\partial i_o} \frac{di_o}{dv} + \frac{\partial f}{\partial v} \right), \text{ and} \\ \frac{\partial d}{\partial v_c} &= FM. \end{aligned} \quad (7)$$

Substituting (6) and (7) into (3), the dc gain of G_c , K , becomes (8), shown at the bottom of the page. The gain K can be made independent of the load current if the terms in parenthesis are zero, i.e. if

$$\frac{\partial f}{\partial v} = R_i \frac{I_o}{V_g}, \quad \text{and} \quad \frac{\partial f}{\partial i_o} = R_i \frac{1}{D'}. \quad (9)$$

Conditions (9) are satisfied if $f = R_i(i_o v)/v_g$, so that the average duty cycle becomes

$$d = -FM \left(R_i i - v_c - R_i \frac{i_o v}{v_g} \right). \quad (10)$$

The modification of the current loop (10) was also proposed in [4], [5] with different motivation. It will be referred to in the text as nonlinear current mode control (NCMC).

In practical realization, there will be some mismatch between the current sensor gain, R_i , and the constant which multiplies the term $(i_o v)/v_g$ in (10). In this case, the average duty cycle is given by:

$$d = -FM \left(R_i i - v_c - R'_i \frac{i_o v}{v_g} \right). \quad (11)$$

If $R'_i = R_i$ the ideal NCMC (10) is obtained, and if $R'_i = 0$, NCMC reduces to CCMC.

III. SMALL-SIGNAL ANALYSIS OF NCMC

A small-signal model of NCMC can be derived by linearizing (1) and (11) at the operating point. In the following, it is assumed that the converter is loaded by a current sink i_o , where $i_o \neq i_o(v)$ is an additional independent input (unterminated analysis [6]). If the input voltage v_g is also variable, the small-signal model is

$$\begin{aligned} \frac{d\hat{i}}{dt} &= -\frac{D'}{L} \hat{v} + \frac{V}{L} \hat{d} + \frac{1}{L} \hat{v}_g \\ \frac{d\hat{v}}{dt} &= \frac{D'}{C} \hat{i} - \frac{I_o}{CD'} \hat{d} - \frac{\hat{i}_o}{C} \end{aligned} \quad (12)$$

$$\hat{d} = \frac{\partial d}{\partial i} \hat{i} + \frac{\partial d}{\partial v} \hat{v} + \frac{\partial d}{\partial v_c} \hat{v}_c + \frac{\partial d}{\partial i_o} \hat{i}_o + \frac{\partial d}{\partial v_g} \hat{v}_g \quad (13)$$

where

$$\begin{aligned} \frac{\partial d}{\partial i} &= -FM R_i, & \frac{\partial d}{\partial v} &= FM R'_i \frac{I_o}{V_g}, \\ \frac{\partial d}{\partial i_o} &= FM R'_i \frac{V}{V_g}, & \frac{\partial d}{\partial v_g} &= -FM R'_i \frac{V I_o}{V_g^2}, \\ \text{and } \frac{\partial d}{\partial v_c} &= FM. \end{aligned} \quad (14)$$

A. Control-to-Output Transfer Function

The NCMC control-to-output transfer function can be obtained by substituting (13) and (14) into (12) with $\hat{i}_o = 0$ and $\hat{v}_g = 0$. In a low and mid-frequency range, G_c is approximately given by:

$$G_c = \frac{\hat{v}}{\hat{v}_c} = K \frac{1 - s/z_G}{1 + s/p_L} \quad (15a)$$

$$K = \frac{FM V_g}{D'^2 + \left(I_o FM R_i - V_g FM \frac{\partial f}{\partial v} \right) + V \frac{di_o}{dv} \left(FM R_i - D' FM \frac{\partial f}{\partial i_o} \right)}. \quad (8)$$

where

$$K = \frac{FMVD'}{D'^2 + I_o FM R_i \left(1 - \frac{R'_i}{R_i}\right)} \quad (15b)$$

$$z_G = \frac{D'^2 V}{L I_o} \quad (15c)$$

and

$$p_L \approx \frac{1}{LC} \frac{D'^2 + FM I_o R_i \left(1 - \frac{R'_i}{R_i}\right)}{FM \left(R_i \frac{V}{L} + R'_i \frac{I_o^2}{CD'V_g}\right)} \quad (15d)$$

The term multiplying the load current I_o in the denominator of the dc gain K given by (15a) contains the factor $(1 - R'_i/R_i)$. For CCMC $R'_i = 0$, and the dependance of K on the load current is pronounced as was shown in Fig. 2. For ideal NCMC $R'_i = R_i$, and $K = FMV/D'$, i.e. the dc gain is independent of the load current, as it was derived in Section II. For practical NCMC $R'_i \neq R_i$, but (R'_i/R_i) is close to one, so that dependance of K on I_o is significantly reduced.

The zero z_G in (15b) is the positive zero (right half plane zero) of a boost converter [2].

The pole p_L is a low-frequency pole obtained by implementing current-mode control. It follows from (15c) that with NCMC, p_L is moved toward lower frequencies. Eq. (15c) was derived assuming that the second pole of (12)-(14), p_H , is much higher than p_L , i.e., $p_H \gg p_L$.

B. Output Impedance

The output impedance $Z_o = -\hat{v}/\hat{i}_o$ can be calculated by substituting (13) and (14) into (12) with $\hat{v}_c = 0$ and $\hat{v}_g = 0$. In a low and mid-frequency it is approximately given by:

$$Z_o = -\frac{\hat{v}}{\hat{i}_o} = Z_{dc} \frac{1 + s/z_I}{1 + s/p_L}, \quad (16a)$$

where

$$Z_{dc} = \frac{FMV(R_i - R'_i)}{D'^2 + I_o FM R_i \left(1 - \frac{R'_i}{R_i}\right)} \quad (16b)$$

$$z_I = \frac{V}{L} FM \frac{R_i - R'_i}{1 + I_o FM R'_i \frac{1}{D'^2}} \quad (16c)$$

and p_L is given by (15d).

From (16b) Z_{dc} is smaller when $R'_i > 0$ (NCMC), than when $R'_i = 0$ (CCMC). However, a small change of R'_i around R_i will change the numerator sign of Z_{dc} . Since the denominator sign does not change simultaneously, the dc value of the output impedance will be positive for $R'_i < R_i$, and it will be negative for $R'_i > R_i$. A similar conclusion holds for the zero z_I given by (16c). Hence, slight change of R'_i around R_i changes the sign of Z_{dc} and moves z_I from the complex left half plane to the right half plane. In this way, the change of R'_i produces qualitative change of the phase of the output impedance, which may be undesirable. A possible solution is to keep R'_i as close as possible to R_i while assuring $R'_i < R_i$.

C. Audio Susceptibility

Audio susceptibility, $A = \hat{v}/\hat{v}_g$, can be derived by substituting (13) and (14) into (12), while setting $\hat{v}_c = 0$ and $\hat{i}_o = 0$. It is approximately given by:

$$A = \frac{\hat{v}}{\hat{v}_g} = \frac{1}{D'} \frac{1 + s/z_A}{1 + s/p_L}. \quad (17a)$$

The zero z_A is

$$z_A = \frac{1}{L} \frac{D' + FM R_i \frac{I_o}{D} \left(1 - \frac{R'_i}{R_i}\right)}{\frac{I_o'}{D} FM R'_i \frac{I_o V}{V_g^2}}. \quad (17b)$$

The dc gain is equal to $1/D'$, and does not depend on R'_i . For $R'_i = 0$ (CCMC), the zero z_A tends to infinity. However, when $R'_i > 0$, the zero z_A appears at some finite frequency which can be below half of the switching frequency. For that reason, audio susceptibility with NCMC may be above the audio characteristic with CCMC at higher frequencies.

IV. COMPLETE SMALL-SIGNAL MODEL OF NCMC

The small-signal model (12), derived from (1) and (11), predicts major features of NCMC. However, it does not model the system accurately because the parasitic resistances are not included in the boost converter model (1), and the sampling effect of the current-mode control is not described by (11). Moreover, if the peak current-mode control (CIC) is employed, the input or output voltage changes have direct effect on the duty cycle and average current [3], which is not covered by (11).

Proper modeling of the sampling effect of current-mode control is presented in [3], and is briefly reviewed in Appendix B. The change of the duty cycle and average current due to the change of the input voltage is modeled by feedforward gain k_f and by the output voltage feedback gain k_r . The small-signal model of CCMC given in [3] can be extended for NCMC by adding the feedback branches caused by the last term in (11).

When parasitic resistances are included, the average duty ratio for NCMC is

$$d = -FM \left(R_i i - v_c - R'_i \frac{i_o v_o}{v_g} \right). \quad (18)$$

where the output voltage v_o is

$$v_o = v + D' R_c i - R_c i_o, \quad (19)$$

and R_c is the parasitic resistance of the output capacitor (ESR). The perturbed duty ratio then is

$$\hat{d} = \hat{d}_o + FM R'_i \left(\frac{V_o}{V_g} \hat{i}_o + \frac{I_o}{V_g} \hat{v}_o - \frac{I_o V_o}{V_g^2} \hat{v}_g \right) \quad (20)$$

where $\hat{d}_o$ is the perturbed duty cycle for CCMC derived in [3].

Block diagram of the complete small-signal model for NCMC is shown in Fig. 3. The transfer functions of different

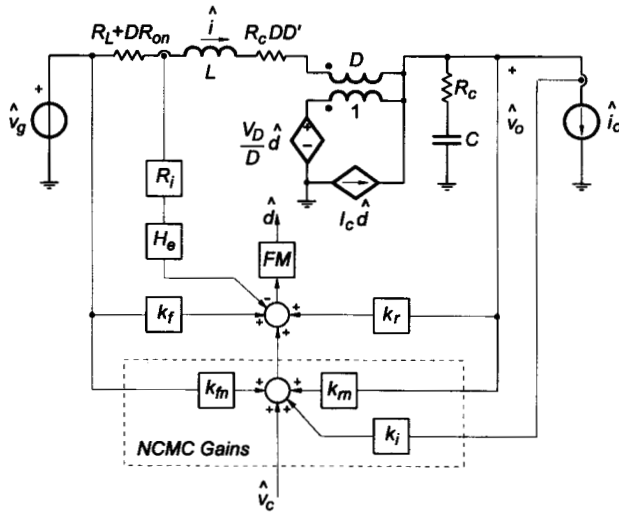


Fig. 3. Complete small-signal model of NCMC.

TABLE I
PARAMETERS OF COMPLETE NCMC SMALL-SIGNAL MODEL

Operating Point	$V = \frac{1}{D'} \left[-(R_L + D R_{on}) \frac{I_o}{D'} + V_s - D R_e I_o \right]$ $V_D = -V - \frac{I_o}{D'} [(D - D') R_e - R_m], \quad I_e = -\frac{I_o}{D'}$
CCMC Gains	$k_f = -\frac{T_s R_L}{2L}, \quad k_r = \frac{D'^2 T_s R_L}{2L}, \quad H_s = 1 - \frac{T_s}{2} s + \frac{T_s^2}{\pi^2} s^2$
NCMC Gains	$k_{fn} = -R'_i \frac{I_o V_o}{V_s^2}, \quad k_{rn} = R'_i \frac{I_o}{V_s}, \quad k_i = R'_i \frac{V_o}{V_s}$

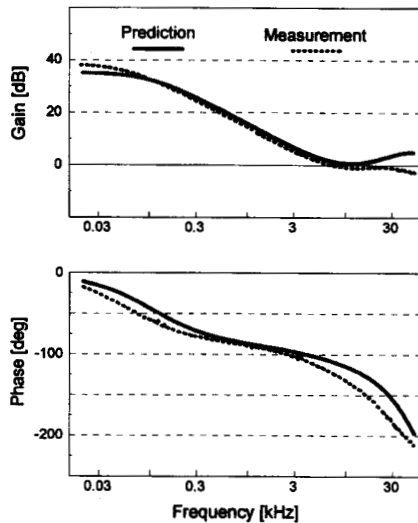


Fig. 4. Measured and predicted NCMC control-to-output transfer function.

blocks in the model are summarized in Table I. The control-to-output transfer function measured for the boost converter with parameters given in Appendix A is compared in Fig. 4 with the transfer function obtained from the above model. Relatively good agreement between the theoretical and experimental results confirms that the model in Fig. 3 can be used for prediction of NCMC small-signal characteristics.

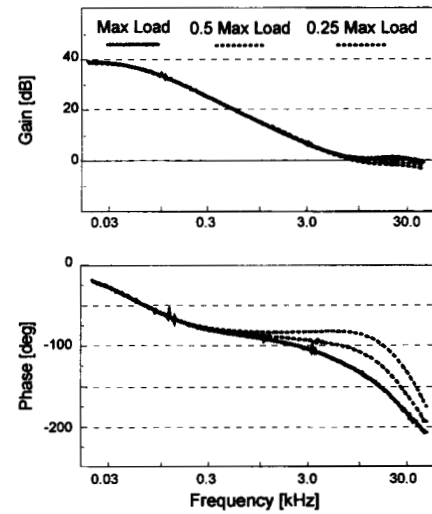


Fig. 5. NCMC control-to-output transfer functions with constant current load.

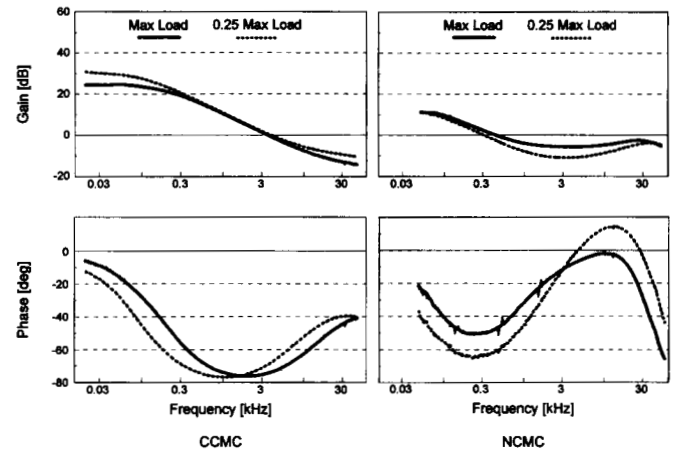


Fig. 6. Output impedance with constant current load.

V. EXPERIMENTAL RESULTS

The proposed control was verified and compared with the conventional one, on an experimental 250-W boost converter prototype. The converter and control parameters are given in Appendix A. CCMC was implemented using standard constant frequency, peak current mode control. For NCMC, the additional nonlinear term $(i_o v_o)/v_g$ was obtained with an analog multiplier/divider and was simply added to the existing CCMC current loop. Load current i_o was measured with a grounded 0.4Ω shunt followed by an op-amp.

A. Small-Signal Characteristics with Open Voltage Loop

The small-signal analysis results derived in Section III were verified first. The converter was loaded with the constant current source, which corresponds to the unterminated analysis of Section III.

Measured NCMC control-to-output transfer functions, G_c , are shown in Fig. 5. As expected, the dc gain does not change with the load current.

Converter output impedances with CCMC and NCMC are shown in Fig. 6. According to (16b), dc value of the output impedance with NCMC depends strongly on the difference

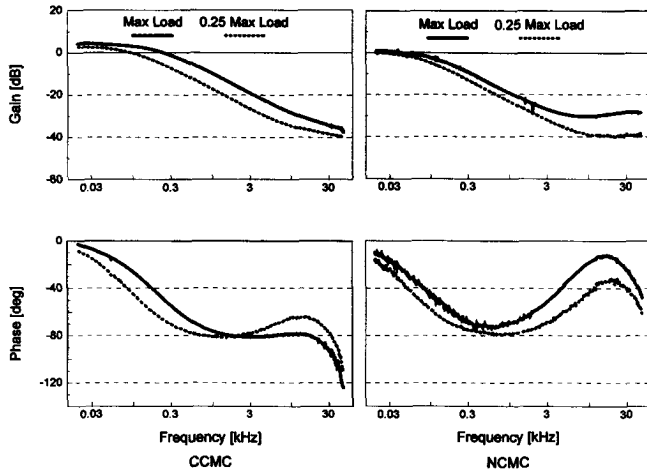


Fig. 7. Audio susceptibility with constant current load.

between R'_i and R_i . In the experiments R'_i was slightly less than R_i so that for all load conditions the output impedance phase was between -90° and $+90^\circ$ at low and middle frequencies.

Audio susceptibility characteristics are shown in Fig. 7. The NCMC audio susceptibility zero z_A in (17b), is below half of the switching frequency, and it causes the characteristic to stay constant in the mid-frequency range, instead of decreasing with -1 slope as with CCMC.

B. Output Voltage Loop Design

It is assumed that the converter is required to operate with both linear resistive load and CPL, and that the load current I_o varies in the range $0.25I_{o\max} < I_o < I_{o\max}$. A voltage compensator should be designed so as to ensure stability in all cases.

NCMC control-to-output transfer functions for both load types are very close to the transfer functions measured with constant current load shown in Fig. 3. Since the transfer functions do not change appreciably, a single optimal voltage compensator can be designed for all load conditions.

To assure stability, the voltage compensator for CCMC must be designed for the case when control-to-output transfer function has the largest gain. The largest gain appears when the CCMC converter operates with CPL and it is very close to the dc gain of NCMC (Fig. 2 and Fig. 5). Thus, the same voltage compensator is used for both, CCMC and NCMC. A relatively conservative compensator was designed for the output voltage loop-gain crossover frequency of 800 Hz and the phase margin of 60° . The compensator transfer function is given in Appendix A.

In the experiments, CPL was generated by a load converter which was implemented as a boost converter with closed voltage loop. The converter and control parameters are given in Appendix C. The load converter input impedance is shown in Fig. 8. The input impedance has -1 slope at higher frequencies due to a small capacitor which was added at the input of the load converter to reduce noise.

Measured output voltage loop gain for NCMC and CCMC for different load conditions is shown in Fig. 9. Since the open

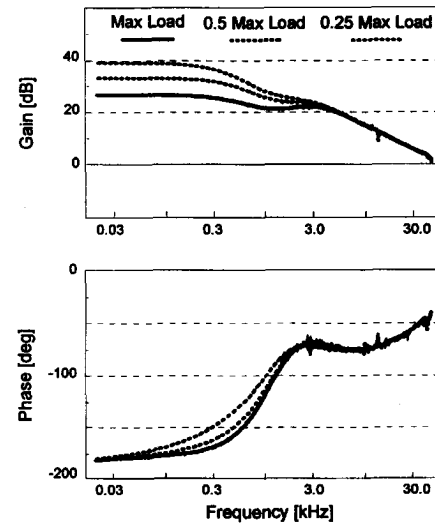


Fig. 8. Load converter input impedance (CPL).

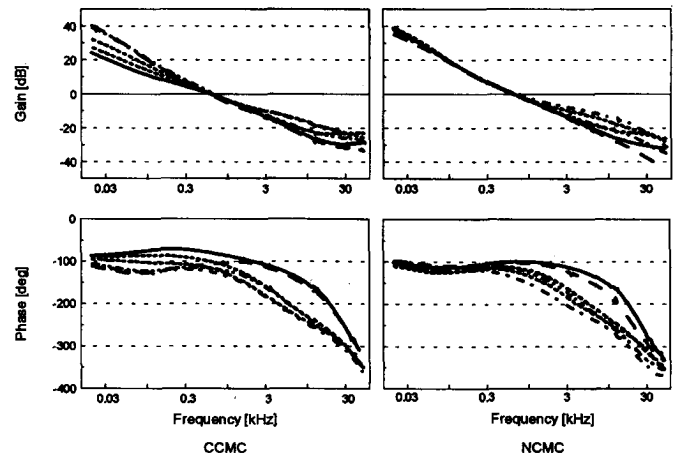


Fig. 9. Output voltage loop-gain variations for linear resistive load and CPL changes between 0.25-1 nominal current.

loop dc gain in CCMC changes with different load conditions, the loop-gain in Fig. 9 varies significantly inside the loop bandwidth. Conversely, the NCMC loop-gain is constant at low and middle frequencies, which illustrates the robustness of this algorithm.

C. Large Signal Response

The 50% to 100% step load response was measured for the boost converter with the closed output voltage loop and both types of control. Fig. 10 shows the load current step and the output voltage response when the converter is operating with linear resistive load. In the case of NCMC, the voltage overshoot is much smaller, and the transient response is faster. Similar conclusion holds for the transient response with CPL which is shown in Fig. 11.

Start-up instability can occur in the boost converter if the output voltage compensator gets saturated [7]. In this work, the effects of the nonlinear control on the potential start-up instability of a boost converter have not been investigated. Instead, the conventional start-up scheme was used. There, the output voltage is first built up to the input voltage level through the diode without turning on the switch. Then, the

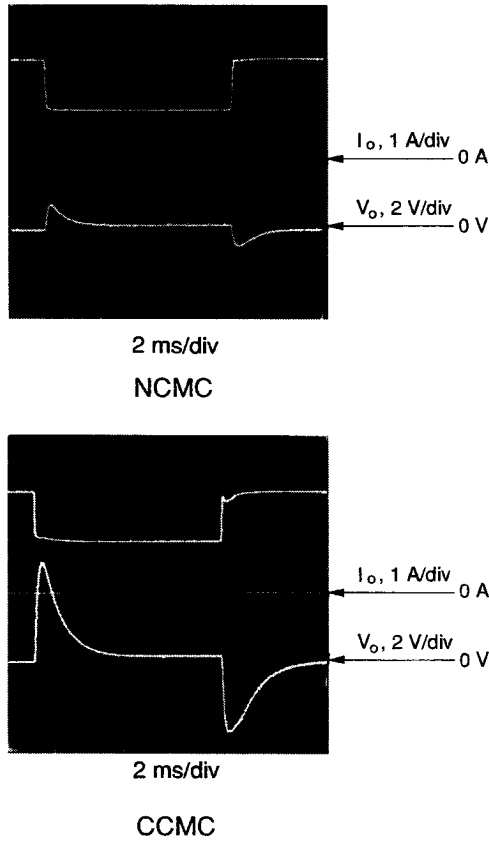


Fig. 10. Step load response with linear resistive load.

output voltage reference is slowly ramped up to the desired value. In this way, the compensator saturation and the potential start-up instability are avoided.

VI. NCMC FOR BUCK AND FLYBACK CONVERTERS

It can be argued that the CPL will have similar effects on the control-to-output transfer functions of the buck and flyback (buck-boost) converters with the CCMC. It was shown in [1] that in the buck converter with CPL the CCMC does not necessarily provide small-signal stability. Modification of the current-mode control can be derived by following the approach from Section II, and in the case of the buck converter, it reduces to the load-current feedforward, i.e. (11) becomes

$$d = -FM(R_i i - v_c - R'_i i_o). \quad (21)$$

Similarly as in the boost converter, the control law (21) practically eliminates dc gain variations and reduces the converter output impedance significantly compared to the output impedance with CCMC. The modification (21) was also derived in [4] and [5].

Since the flyback (Fig. 12) and the boost converters exhibit very similar small-signal dynamics (e.g. RHP zero in the control-to-output transfer function), the results obtained for the boost can be applied to the flyback, and the modification of the current-loop (11) will provide invariant dc gain. However, using the approach from [4] and [5], the control law

$$d = -FM \left(R_i i - v_c - R'_i \frac{v - v_g}{v_g} i_o \right) \quad (22)$$

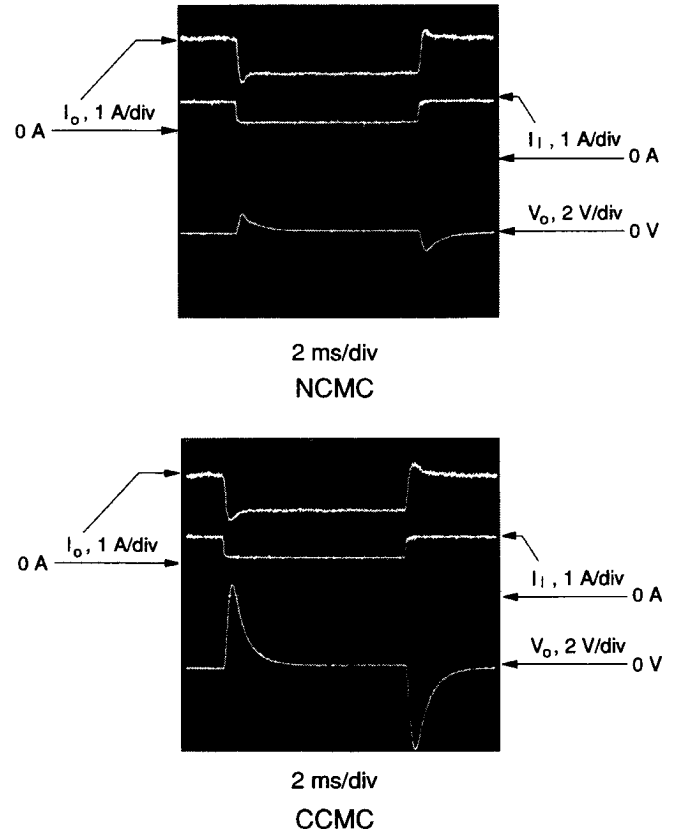


Fig. 11. Step load response with cpl.

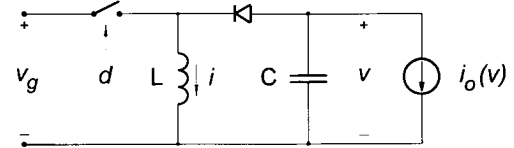


Fig. 12. Flyback (buck-boost) converter.

is obtained. Equation (22) will also eliminate dc gain variations, and will additionally reduce the audio susceptibility compared to the NCMC given by (11).

VII. CONCLUSIONS

The standard current loop of a boost dc-dc converter is modified in order to eliminate closed loop sensitivity to the nature and magnitude of resistive load. The modification is derived from the requirement that the dc gain of the control-to-output transfer function with closed current loop remain constant for load changes. With the alternative formulation presented in [4] and [5], this can be achieved if the current reference is calculated from the equality of input and output powers, i.e.

$$v_g i_{ref} = v_o i_o \rightarrow i_{ref} = \frac{v_o i_o}{v_g}. \quad (23)$$

If this current reference is used in the current loop (2), the duty cycle becomes

$$\begin{aligned} d &= -FM [R_i (i - i_{ref}) - v_c] \\ &= -FM \left[R_i i - R_i \frac{i_o v_o}{v_g} - v_c \right] \end{aligned} \quad (24)$$

which is the same as (10).

The above formulation provides for a good insight into the properties of NCMC.

- 1) Since the reference current (23) is derived from the power balance, the effect it introduces into the feedback loop of NCMC is similar to the one CPL imposes on the CCMC converter externally. This explains why the NCMC control-to-output transfer function (Fig. 5) is the same as the CCMC transfer function with CPL (Fig. 2). Both do not depend on the load level. The added advantage of NCMC is that the transfer function remains approximately invariant even with different resistive loads.
- 2) The information about the load current level is fed directly into the current loop through the reference current term, making the dc output impedance practically zero, even without closing the output voltage loop. This is obvious from (24) because any change in i_o will immediately change the duty cycle, which will result in the adjustment of the inductor current i to the new reference level. The same effect results in significantly reduced overshoot and response time to large load step changes.

Two points should be noted in connection with power balance formulation. First, the input and output powers are equal only in steady-state, even for an ideal converter, because of the energy storage components within the converter. Second, there are always additional losses in the converter. As a result, even when $R_i = R'_i$, the effects of the NCMC are pronounced only at low frequencies, the dc gain is not completely invariant, and the dc output impedance is not exactly equal to zero. These points were not considered in [4] and [5].

The NCMC provides significant performance improvement over CCMC and is easy to implement. It requires only additional sensing of the input voltage and load current and a multiplier.

APPENDIX A

EXPERIMENTAL BOOST CONVERTER PARAMETERS

Power stage: $V_g = 40$ V, $V_o = 80$ V, $I_{o\max} = 3.2$ A, $f_s = 100$ kHz, $L = 90$ μ H, $C = 51$ μ F, and $R_c = 0.25$ Ω .

Current loop: $R_i = 0.23$ Ω , $R'_i = 0.21$ Ω , and $FM = 0.62$ V⁻¹.

Voltage compensator:

$$H_v = \frac{300(1 + s/5455)}{s(1 + s/102167)}.$$

APPENDIX B

CONTINUOUS-TIME SMALL-SIGNAL MODEL FOR CURRENT-MODE CONTROL [3]

The block diagram of a boost converter with CCMC is shown in Fig. 3, if the block entitled "NCMC gains" is excluded. H_e represents the sampling action of the current-mode control, k_f and k_e provide feedforward of the input and feedback of the output voltage, FM is the modulator gain, $FM = 1/((S_e + S_n)T_s)$, and S_n is the on-time slope.

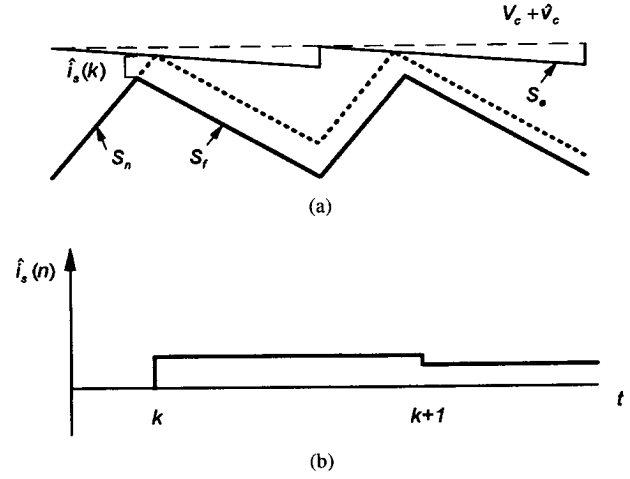


Fig. 13. Current-mode control modulator waveforms, (a) control voltage perturbation, and (b) sampled current perturbation.

To derive H_e , a perturbation is introduced in the control voltage, $\hat{v}_c$ (Fig. 13). From Fig. 13 the closed current-loop response to $\hat{v}_c$ can be obtained as

$$\hat{i}(k+1) = -\alpha\hat{i}(k) + \frac{1}{R_i}(1 + \alpha)\hat{v}_c(k+1) \quad (B1)$$

where $\alpha = (S_f - S_e)/(S_n - S_e)$, and S_f is the off-time slope. From (B1) it follows

$$H(z) = \frac{\hat{i}(z)}{\hat{v}_c(z)} = -\frac{1}{R_i}(1 + \alpha)\frac{z}{z + \alpha}. \quad (B2)$$

The continuous-time representation of (B2) is

$$H(s) = \frac{\hat{i}(s)}{\hat{v}_c(s)} = -\frac{1}{R_i} \frac{1 + \alpha \exp(sT_s) - 1}{sT_s \exp(sT_s) + \alpha}. \quad (B3)$$

H_e can be deduced from (B3) and Fig. 3 by comparison as

$$H(s) = \frac{sT_s}{\exp(sT_s) - 1} \quad (B4)$$

which can be further simplified to

$$H_e(s) \approx 1 + \frac{s}{\omega_n Q_z} + \frac{s^2}{\omega^2} \quad (B5)$$

where $\omega_n = \pi/T_s$ and $Q_z = -2/\pi$.

The describing function for the inductor current in terms of the control voltage and operating conditions for the trailing-edge modulation is

$$R_i\langle i \rangle = V_c - DT_s S_e - \frac{S_f D' T_s}{2}. \quad (B6)$$

The feedforward gain, k_f , can be deduced by comparing $\hat{i}/\hat{v}_g = \partial\langle i \rangle/\partial V_g$ and $\hat{i}/\hat{v}_g$ derived from the circuit in Fig. 3 (without NCMC gains) for steady-state conditions.

The feedback gain, k_e , can be deduced by comparison of $\hat{i}/\hat{v}_o$ expressions derived from (B6) and from the circuit in Fig. 3.

APPENDIX C LOAD CONVERTER PARAMETERS

Power stage: $V_g = 80$ V, $V_o = 133$ V, $I_{o\max} = 1.9$ A, $f_s = 100$ kHz, $L = 400$ μ H, $C = 57$ μ F, and $R_c = 0.25$ Ω .

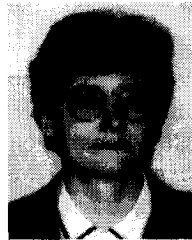
Current loop: $R_i = 0.34$ Ω , and $FM = 0.55$ V⁻¹.

Voltage compensator:

$$H_v = \frac{45(1 + s/4517)}{s(1 + s/82000)}.$$

REFERENCES

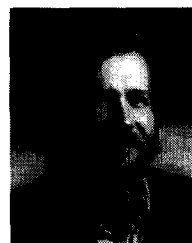
- [1] B. Cho and B. Choi, "Analysis and design of multi-stage distributed power systems," in *Intelec'91*, pp. 220-226, 1991.
- [2] V. Vorperian, "Simplified analysis of PWM converters using the model of the PWM switch, Part I: Continuous conduction mode," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 26, no. 3, pp. 490-497, 1990. pp. 1-9, 1989.
- [3] R. B. Ridley, "A new, continuous-time model for current-mode control," *IEEE Trans. Power Electron.*, vol. 6, no. 2, pp. 271-280, 1991.
- [4] R. Redl and N. O. Sokal, "Optimizing dynamic behavior with input and output feedforward and current-mode control," in *Proc. Powercon7*, pp. H1-1-H1-16, 1980.
- [5] N. O. Sokal and R. Redl, "Near-optimum dynamic regulation of dc-dc converters using feed-forward of output current and input voltage with current-mode control," *IEEE Trans. Power Electron.*, vol. 1, no. 3, pp. 181-192, July 1986.
- [6] B. Cho, "Modeling and analysis of spacecraft power systems," Ph.D. Dissertation, Blacksburg, VA: VPI & SU, 1985.
- [7] R. W. Ericson, S. Cuk, and R. D. Middlebrook, "Large-signal modeling and analysis of switching regulators," in *IEEE PESC'82, Rec.*, pp. 240-250, 1982.



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